

CSC 380 — Midterm Study Guide (PROVISIONAL · LIGHT · date NV)

Start Here

Guide updated: 2026-09-24 afternoon (America/Phoenix) -- Start Here rewrite. **LIGHT / date NV.** Reviewed 2026-09-24 evening by Claude Opus 5.5 (see §8c).

If you only read one thing: do nothing midterm-specific until after Oct 8, except Quiz 04 practice. Re-verify the **tentative** Oct 20 date during Oct 13-19. If it is confirmed, use the contingent plan below. If not, keep it light.

Next scheduled assessment

Field	Value	Label
Assessment	CSC 380 Midterm	item exists
Date/time	Syllabus says tentatively Tue Oct 20, 2026, 5:00-6:15 PM in regular lecture	date-NV -- do not treat as fixed
Format	Not published. Gradescope delivery unconfirmed	NV
Coverage	Syllabus weeks 1-8: pandas, viz, probability, RVs, special distributions, covariance/correlation, review	PROVISIONAL -- schedule-based
Not yet lectured	Special distributions (wk 6), covariance/correlation (wk 7), midterm review (wk 8)	study only as lectured

Do **not** schedule study during class blocks (Tue/Thu through Dec 9, Arizona time): CSC 252 9:30-10:45 AM; CSC 335 12:30-1:45 PM; CSC 345 2:00-3:15 PM; CSC 337 3:30-4:45 PM; CSC 380 5:00-6:15 PM. All times below are **Arizona (MST, UTC-7)**. Combined daily order: `weekly/2026-09-24-weekly-study-order.md`.

Do not densify until the instructor confirms the date (re-check D2L / xinchenyu calendar during Oct 13-19). Nearer work: Quiz 04 practice (scope confirmed, due NV), HW03 Sep 29 11:59 PM, HW4 Oct 13 11:59 PM (when released). Finals horizon (not this 14-day window): Wed Dec 16 6:00-8:00 PM.

Topics to study first (order + WHY)

1. **Quiz 04 published subset (380-004-*)** -- WHY: Bayes / LOTP / independence is the only **confirmed** midterm-overlapping slice; use the Quiz 04 guide.
2. **380-L-prob1 + 380-L-comb** -- **conditional probability + combinations** -- WHY: weeks 2-3 already lectured; combinations was missing from older drafts.
3. **380-L-pandas + 380-L-viz** -- WHY: week 1; short recall, not a marathon.
4. **380-L-rv then 380-L-special / 380-L-corr** -- WHY: weeks 4-7; add only after lecture.

5. **Do not start** modeling / regression / classification (post-midterm).

Confirmed vs schedule-based: Nothing about the midterm date or blueprint is confirmed. Weeks 1-8 are a syllabus timeline, not a published scope.

Day-by-day tasks (LIGHT; no class-block study)

When	Window	Task	Guide / slides / problems	Min
Now - Oct 1	After HW03 / 252 Test 2 only	Quiz 04 practice. Not a midterm marathon.	Quiz 04 guide	see Quiz 04
Fri Oct 9	No class; after the Oct 8 exam cluster	One light pass over the 380-L-* map + L1-L10. (Moved from Sat Oct 3, which is the heaviest 337/335/345 prep day.)	Sec 2-4	35
Mon Oct 12	Anytime	Aim to finish HW4 written + Code tonight, if released. Tue Oct 13 is a class day that also carries CSC 345 PP2 (2:00 PM) and 252 HW3 (7:00 PM, rule-derived).	HW4	HW4
Tue Oct 13	HW4 due 11:59 PM (written + Code). Free gaps: before 9:30, 10:45-12:30, after 6:15	Submit HW4. Then re-verify midterm date/scope.	D2L / https://xinchenyu.github.io/csc380/calendar/	10
Oct 13-19	Outside class blocks	Re-verify date. Densify only if confirmed .	--	--
If Oct 20 is confirmed	Sit 5:00-6:15 PM. Night before: calm review only	L1-L11 + Quiz 04 fluency	30 light	

Contingent plan -- use ONLY if the Oct 20 date is confirmed by about Tue Oct 13.

This adds no new date; it only says how to use the week if the tentative date holds. Thu Oct 15 is CSC 252 Test 3, so the deep days are after it:

When (contingent)	Task	Min
Fri Oct 16	pandas + viz recall (L8, L9); random variables (L3, L4)	60
Sat Oct 17	Weeks 4-7 as lectured : distributions, covariance/correlation (L5); redo Quiz 04 Mock C and D	90
Sun Oct 18	60-minute paper mock: L1-L11 plus two Quiz 04 mocks; then redo misses	80
Mon Oct 19	Light review of misses only	30
Tue Oct 20	Calm buffer before 9:30 or 10:45-12:30 only; exam 5:00-6:15 if confirmed	10

Is this enough time? The light plan (~1.5 hours before Oct 13) is right while the date is NV. If the date is confirmed, the contingent plan adds about 4.5 hours, which is adequate if it is a lecture-period (5:00-6:15) midterm as tentatively listed.

Final self-check (light)

- ☐ Weeks 1-8 concept map reviewed
- ☐ Quiz 04 topics solid
- ☐ L1-L11 correct
- ☐ Date and scope re-verified during Oct 13-19 **before** densifying
- ☐ No regression/NN early study
- ☐ No invented Tuesday quiz dates

Slide excerpts

Reuse the Quiz 04 excerpts (Probability 2 short p.19 LOTP and p.47 Bayes) under `csc380/assessments/excerpts/`. For pandas/viz, review `csc380-26f380_data1.pdf` on the harvest path -- no extra excerpt pulled (keep this guide light).

Banner — date NEEDS VERIFICATION; scope PROVISIONAL; LIGHT guide. - **Date:** the syllabus says "**tentatively**" Tue Oct 20, 2026, 5:00–6:15 PM Arizona time, in the regular lecture. The calendar also lists Oct 20. The date stays **Needs verification** until the instructor confirms it. - **Scope:** coverage is **not published**. It is provisional, from syllabus weeks 1–8: pandas, visualization, probability, random variables and distributions, covariance and correlation, and midterm review. - **Light only:** this is a concept map plus a small practice set. Do not do dense prep that assumes a confirmed date.

Canonical maps: `csc380-topic-map.md` · `announced-assessments-inventory-2026-09-24.md` · `site-audit-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00–6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Midterm (light provisional guide)	Item exists

Field	Value	Status
Date/time	Tentatively Tue Oct 20, 2026, 5:00–6:15 PM Arizona	Needs verification
Format	Not published. Gradescope delivery is not confirmed.	NV
Coverage	Syllabus weeks 1–8	PROVISIONAL
Not yet lectured	Special Distributions (week 6), Covariance/Correlation (week 7), and Midterm Review (week 8) are still ahead	Expected
Guide updated	2026-09-24 (America/Phoenix) — Opus + Content-fold (light)	

All times are **Arizona time (MST)**.

Sources (Content-pass 2026-09-24)

Source	Role
Content Resources/quiz01_03.pdf	Early quiz-style practice grain (descriptive stats; sample space; LOTP+Bayes). Overlaps Quiz 04 published subset.
Content Homework/homework01.zip, homework02.pdf, homework03.pdf	HW practice context (pandas/viz; probability; Bayes/independence). Not midterm solutions.
Syllabus weeks 1–8	PROVISIONAL midterm coverage map

2. Topic checklist (light concept map)

Order	Section ID	Topic (syllabus week — provisional)
1	380-L-intro	Course overview (wk 1)
2	380-L-pandas	Basic data manipulation with pandas (wk 1)
3	380-L-viz	Basic visualization (wk 1)
4	380-L-prob1	Probability intro; conditional probability (wk 2)
5	380-L-comb	Combinations (wk 3) — <i>new in this review</i>
6	380-Q04-indep	Independence (wk 3; also Quiz 04)
7	380-Q04-bayes / 380-Q04-lotp	Bayes and LOTP (Quiz 04 published subset)
8	380-L-rv	Random variables; distributions; discrete vs continuous (wk 4–5)
9	380-L-special	Special distributions (wk 6 — as lectured)
10	380-L-corr	Covariance and correlation (wk 7 — as lectured)
11	380-L-review	Midterm review (wk 8)

Do **not** start the post-midterm material (modeling, regression, classification).

3. Mini explanations

- **pandas.** These are standard pandas; match your class notebooks.
- Filter rows: `df[df["age"] > 30]`
- Select columns: `df[["a", "b"]]`
- Group means: `df.groupby("team")["score"].mean()`
- Missing values: `df.isna().sum()` and `df.dropna()`
- **Chart choice.**

What you want to show	Chart
Distribution of one numeric variable	Histogram or box plot
Two numeric variables	Scatter plot
Comparison across categories	Bar chart
Change over time	Line chart

- **Combinations.** $C(n, k) = n! / (k!(n-k)!)$ counts **unordered** selections. Permutations $P(n, k) = n! / (n-k)!$ count **ordered** ones.
- **Random variables.** For a discrete X , $E[X] = \sum x \cdot P(x)$. $\text{Var}(X) = E[X^2] - (E[X])^2$.
- **Special distributions** (as lectured).
- Bernoulli(p): mean p , variance $p(1-p)$.
- Binomial(n, p): mean np , variance $np(1-p)$.
- Uniform, Poisson, Normal, and Exponential follow if covered.
- **Covariance and correlation.** $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$. $\text{Corr} = \text{Cov} / (\sigma_X \cdot \sigma_Y)$, which always lies in $[-1, 1]$. Correlation measures **linear** association only. Check which denominator the course uses for sample covariance (n or $n - 1$).
- **Bayes, LOTP, independence.** These are the published Quiz 04 topics; see the Quiz 04 Practice Guide.

4. Light practice (self-contained)

L1 (recall). Define independence, and write correlation in terms of covariance.

L2 (combinations). How many 3-person committees can be chosen from 7 people? What is the probability of exactly 2 heads in 4 fair coin flips?

L3 (random variable). For a fair six-sided die X , find $E[X]$ and $\text{Var}(X)$.

L4 (binomial). $X \sim \text{Binomial}(10, 0.3)$. Find $E[X]$ and $\text{Var}(X)$.

L5 (covariance/correlation). For $X = [1, 2, 3]$ and $Y = [2, 4, 6]$, find the sample covariance (denominator $n - 1$) and the correlation.

L6 (Bayes). $P(H) = 0.2$, $P(E | H) = 0.9$, $P(E | \text{not-}H) = 0.3$. Find $P(H | E)$.

L7 (independence). $P(A) = 0.25$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.1$. Are A and B independent?

L8 (visualization). Pick a chart for (a) the distribution of house prices, (b) price vs. square footage, (c) average price by neighborhood.

L9 (pandas). Write one line that gives the mean `score` for each `team` in DataFrame `df`.

L10 (planning). Why must the tentative Oct 20 date be re-verified before making a dense plan?

L11 (Content grain — Practice recommendation). Two labs: Lab A has 90 students (30% prefer Python); Lab B has 60 students (70% prefer Python). (a) $P(\text{prefer Python})$. (b) Given prefer-Python, $P(\text{Lab B})$. (Cite Content quiz01_03 Quiz-03 style; numbers changed.)

4b. Format reminder — Practice recommendation

The Gradescope format notes show that quizzes are about 1.5 points on a single page, and each homework has **two** items (written and Code). The **midterm format is not published**. For practice, write L2-L7 by hand, one per half-page, with every step shown. This is a generic habit, not a claimed midterm format.

5. ANSWER KEY

L1. Independence means $P(A \text{ and } B) = P(A) \cdot P(B)$. $\text{Corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y)$.

L2. $C(7, 3) = 35$. $P(\text{exactly 2 heads in 4 flips}) = C(4, 2)/2^4 = 6/16 = \mathbf{0.375}$.

L3. $E[X] = 3.5$. $E[X^2] = 91/6$. $\text{Var}(X) = 91/6 - 12.25 = \mathbf{35/12} \approx \mathbf{2.917}$.

L4. $E[X] = 10 \cdot 0.3 = \mathbf{3}$. $\text{Var}(X) = 10 \cdot 0.3 \cdot 0.7 = \mathbf{2.1}$.

L5. The means are 2 and 4. The deviations are (-1, 0, 1) for X and (-2, 0, 2) for Y, so the sum of products is $2 + 0 + 2 = 4$. Sample covariance = $4 / (3 - 1) = \mathbf{2}$. The sample standard deviations are 1 and 2, so correlation = $2 / (1 \cdot 2) = \mathbf{1}$, a perfect positive linear relationship. With the population denominator, the covariance is $4/3$, and the correlation is still 1.

L6. $P(E) = 0.9 \cdot 0.2 + 0.3 \cdot 0.8 = 0.42$. $P(H | E) = 0.18 / 0.42 = \mathbf{3/7} \approx \mathbf{0.429}$.

L7. $0.25 \times 0.4 = 0.1 = P(A \text{ and } B)$. **Yes, independent.**

L8. (a) Histogram (or box plot). (b) Scatter plot. (c) Bar chart.

L9. `df.groupby("team")["score"].mean()`

L10. The syllabus marks the date as "tentative". The context rule is never to treat an unconfirmed date as fixed, so re-check on D2L, the course site, or in class during Oct 13-19 before planning around it.

L11. (a) $0.3(90/150) + 0.7(60/150) = 0.3 \cdot 0.6 + 0.7 \cdot 0.4 = 0.18 + 0.28 = \mathbf{0.46}$. (b) $0.28/0.46 = \mathbf{14/23} \approx \mathbf{0.609}$.

6. Common mistakes + readiness (light)

- Studying regression or neural networks early.
- Treating the tentative Oct 20 date as fixed.
- Skipping Quiz 04 fluency, since Bayes, LOTP, and independence are the only **published** subset.
- Mixing up combinations (unordered) with permutations (ordered).
- Reading a correlation near 0 as "no relationship", when it only means no **linear** relationship.
- Inventing quiz dates from the Tuesday cadence.

Ready (light): - [] Weeks 1-8 concept map reviewed - [] Quiz 04 topics solid - [] L1-L11 correct - [] Date and scope re-verified during Oct 13-19 **before** densifying

Start Here (top of this guide) is the canonical day-by-day. The later prep table is kept as a short copy.

7. Prep plan (light — no day-before cram that assumes a confirmed date)

Window	Action
Now – early Oct	Quiz 04 practice and HW3/HW4 probability. Not a midterm marathon.
Fri Oct 9	One light pass over the 380-L-* concept map and L1-L10 (after the Oct 8 exam cluster)
Tue Oct 13	HW4 due 11:59 PM (written plus Code); aim to finish Mon Oct 12 night
Oct 13-19	Re-verify the date and scope on D2L, the site, or the tracker. Densify only after that.
If Oct 20 is confirmed	Use the contingent plan in Start Here (Fri Oct 16 - Mon Oct 19). Sit the midterm at 5:00–6:15 PM. The night before, calm review only.

Do deeper work Mon/Wed/Fri and weekends; keep Tuesdays and Thursdays light.

8. Changes in this review (2026-09-24)

- Kept the **date NV** and **LIGHT** banners unchanged in meaning. Changed "PT" to Arizona time.
- Made the answer key self-contained.** The old key said "see the Quiz 04 guide" for Bayes; all L-items now have worked answers.
- Added the week-3 **Combinations** topic, which was merged into independence and had no practice. Added practice on random variables, the binomial distribution, covariance/correlation, visualization, and pandas (L2–L5, L8–L9).
- Noted that the weeks 6–8 lectures have not happened yet.
- Kept the new practice to ten short items so the guide stays light.

8b. Content-fold (2026-09-24, America/Phoenix)

- Light only: cited Content quiz01_03 + HW01-03; added L11 similar-style LOTP+Bayes drill. Date/scope banners unchanged (NV / PROVISIONAL / LIGHT). No dense invent.

8c. Opus 5.5 review (2026-09-24 evening, America/Phoenix)

- Recomputed L1-L11 (combinations, die mean/variance = 35/12, binomial, sample covariance/correlation, Bayes, independence, the L11 LOTP). **No wrong answers found.**
- Order fix:** the one light 380 pass moved from Sat Oct 3, the heaviest 337/335/345 prep day, to Fri Oct 9, after the exam cluster. The midterm is Oct 20 at the earliest, and that date is tentative.
- Added a Mon Oct 12 HW4 target, because Tue Oct 13 is a class day that also carries 345 PP2 and 252 HW3.
- Added a clearly labeled **contingent** plan (Fri Oct 16 - Tue Oct 20) to use **only if** the tentative Oct 20 date is confirmed. **No date was invented or promoted;** the midterm

is still **date-NV**.

- Converted the remaining non-ASCII math symbols to ASCII, to match the Quiz 04 guide's PDF-safe convention.

9. Disclaimer

Study aid only — not for submitted work. Follow CSC 380 and UA policies. The title marks provisional/NV status on purpose.